2 ALARM DISPLAY AND CORRESPONDING COUNTERMEASURES

Number	Alarm type
000 to 299	Program or setting alarm
300 to 399, 401	Pulse coder alarm
400 to 499 (except 401)	Servo alarm
500 to 599	Overtravel alarm
–	System alarm or I/O Link alarm

Program or setting alarms (PS alarms)

No.	LED display	Description	Countermeasure
000	0	A parameter that requires power–down has been specified.	Turn the power off, then back on.
011	6	The specified feedrate is zero.	Check the feedrate parameter specified with a function code.
013	6	The specified feedrate (maximum feedrate) is zero.	Check the value of parameter No.043, which indicates the maximum feedrate that can be specified.
070	7	More than 32 blocks have been registered for a buffering operation.	Reduce the number of registered blocks to 32.
085	8	Input from the reader/punch interface or the like caused an overrun, parity, or framing error.	Correct the baud rate of the input/output unit (always 4800) or other settings.
086	8	Input from the reader/punch interface or the like includes an input/output unit operation ready signal (DR) that is set to off.	Turn the reader/punch on. Check the cable connection.
087	8	After input from the reader/punch interface or the like stops, character input does not stop even though ten characters have been input.	Check the cable connection.
090	9	Reference position setting cannot be executed normally.	Move the tool in the direction of reference position return in jog mode at a speed that causes the servo position error to exceed 128. Then, specify another reference position setting.
093	0	A first to third reference position return cannot be executed because the reference position has not yet been established.	Set the reference position.
224	0	The reference position has not yet been established. This occurs only when the ZRTN bit of parameter No.001 is set to 0.	Set the reference position.
250	0	Input data 1 is invalid.	Check input data 1, specified with a function code.
251	0	Input data 2 is invalid.	Check input data 2, specified with a function code.
254	0	A function code or mode is invalid.	Check the command code, specified with a function code. Check the mode.
255	0	Operation cannot be activated because an invalid mode is specified or because block execution is in progress.	Check the mode. Check whether a block is being executed.

No.	LED display	Description	Countermeasure
290	0	The interface switch signal (DRC) was switched during block execution.	Switch the signal after block execution stops.
291	0	The speed of an axial movement specified by an external pulse has exceeded the upper limit. This occurs only when the EPEXA bit of parameter No.001 is set to 1.	Check the speed specified by the external pulse. Check the magnification of the external pulse (parameters No.062 and 063).
292	8	A checksum error for the battery–powered memory was detected.	Parameters are cleared. Set the parameters again. If this alarm subsequently recurs, replace the unit.

Pulse coder alarms

No.	LED display	Description	Countermeasure
300	9	A communication error (DTER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder may be defective. Turn the power off. If the alarm recurs when the power is reapplied, replace the serial pulse coder, together with the motor.
301	9	A communication error (CRCER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder or servo amplifier unit may be defective. This error can also be caused by external noise. See the chapter on noise reduction in “Descriptions.”
302	9	A communication error (STBER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder or servo amplifier unit may be defective. This error can also be caused by external noise. See the chapter on noise reduction in “Descriptions.”
303	6	An LED disconnection (LDAL) was detected in the serial pulse coder.	Turn the power off. If this alarm recurs when the power is reapplied, replace the motor.
304	6	A mispulse alarm (PMAL) for the serial pulse coder was detected.	Turn the power off. If this alarm recurs when the power is reapplied, replace the motor.
305	6	A miscount alarm (CMAL) for the serial pulse coder was detected.	Turn the power off. If the alarm recurs when the power is re–applied, replace the motor. If the alarm does not recur, restart the operation from reference position return.
306	8	The motor has overheated (OHAL).	This alarm is issued when the motor has overheated, causing the thermostat to trip. Possible causes include an excessively high ambient temperature and excessively strict operating conditions. Check the actual cause. If it occurs again when the motor is cooled, the motor or servo amplifier may have failed. Replace the faulty motor or servo amplifier.
308	6	A soft phase alarm (SPHAL) was detected.	Turn the power off. This alarm may be caused by noise.
319	8	When the absolute pulse coder is used, the motor has not yet rotated through more than one turn after the first power–up.	Cause the motor to rotate through more than one turn in jog feed mode, then turn the power off then back on.
350	2	The battery voltage of the absolute pulse coder is low.	Replace the battery. Restart the operation from reference position return.

No.	LED display	Description	Countermeasure
351		The battery voltage of the absolute pulse coder is low. (warning)	Replace the battery.
401		A communication error was detected in the serial pulse coder.	Check the continuity of the pulse coder signal cable. If the cable is normal, the pulse coder or servo amplifier unit may be defective. This error can also be caused by external noise. See the chapter on noise reduction in “Descriptions.”

Servo alarms

No.	LED display	Description	Countermeasure
400		The servo motor has overheated (estimated value).	The motor operation condition may be too severe. Check the operation condition.
403		SVU–12, SVU–20: The cooling fins have overheated. (hardware detection)	The load on the motor may be too high. Re–examine the load conditions.
		SVU–40, SVU–80: This alarm will not be issued.	
404		The regenerative discharge unit has overheated.	This alarm is issued when the average regenerative discharge energy is too high (when the acceleration/deceleration frequency is too high, for example).
		(1) When the separate regenerative discharge resistor is not used: For the SVU–12 and the SVU–20, check whether the CX11–6 connector is short–circuited with a dummy connector; for the SVU–40 and the SVU–80, check whether the CX20 and CX23 connectors are short–circuited with a dummy connector. (2) The average regenerative discharge energy may be too high. Decrease the acceleration/deceleration frequency. (3) The separate regenerative discharge unit may not be connected properly. Check the connection. (4) The thermostat of the separate regenerative discharge unit may be defective. Disconnect the separate regenerative discharge unit, then check the thermostat. If the thermostat is open even through the separate regenerative discharge unit is cool, replace the separate regenerative discharge unit. (5) The resistor of the separate regenerative discharge unit may be defective. Disconnect the separate regenerative discharge unit, then check the resistance. If it does not fall in the range of the predetermined resistance ±20%, replace the separate regenerative discharge unit. (6) If (1) to (5) are not the cause of the alarm, replace the servo amplifier unit.	
405		Reference position return could not be executed correctly.	Re–execute reference position return.
		If a value in the range of 4 to 96 is set for parameter No. 032 (CMR), an alarm may be issued. In this case, prevent an alarm from being issued by setting N405 (bit 4 of parameter No. 001) to “1”.	
410		The servo position error in the stop state is larger than the value specified in parameter No.110.	Determine the mechanical cause of the large position error. If no mechanical cause is found, specify a larger value for the parameter.
411		The servo position error during movement is larger than the value specified in parameter No.182.	Determine the mechanical cause of the large position error. If no mechanical cause is found, apply any of the following countermeasures: • Specify a larger value for the parameter. • Specify a lower feedrate. • Increase the time constants.

No.	LED display	Description	Countermeasure
412	0	[SVU–12, SVU–20] An overcurrent alarm is issued.	This alarm is issued when an excessively large current flows in the main circuit.
		(1) Check whether a valid motor number is specified in parameter No.30. (2) Check whether the standard values (see Appendix B) are specified in the current control parameters for servo control. Correct current control is possible only when the standard values are specified for the following parameters: No.70, 71, 72, 78, 79, 84, 85, 86, 87, 88, 89, 90 (3) Disconnect the power line from the servo amplifier unit connector. Then, release the emergency stop state. – If the overcurrent alarm continues to be issued, replace the servo amplifier unit. – If no overcurrent alarm is issued, go to (4). (4) Check the insulation between the ground and each of U, V, and W. If the insulation is satisfactory, go to (5). → If a short–circuit is detected, disconnect the power line from the motor connector. Then, check the insulation between the ground and each of U, V, and W of the motor. → If a short–circuit is found between the ground and U, V, or W of the motor, replace the motor. → If the insulation is satisfactory, replace the power line. (5) Connect the power line. Observe the waveform of the motor current (IR, IS) while the motor is accelerating or decelerating. See the description of the checking of the motor current in Appendix E, "Servo Check Board". If the motor current (IR, IS) does not exhibit a normal sine wave, replace the servo amplifier unit. (6) Check if the motor current (IR, IS) waveforms include noise. → When noise is included, take an action such as making a connection to shield ground. → When noise is not included, replace the servo amplifier unit. (7) If (1) to (4) above are not the cause of the alarm, the pulse coder, command cable, or internal hardware of the CNC may be defective.	
		[SVU–40, SVU–80] An overcurrent alarm or IPM alarm is issued.	This alarm is issued in the following cases: • This alarm is issued when an excessively large current flows in the main circuit. • This alarm is issued when an error (overcurrent, overheat, low IPM control power supply voltage) is detected in the IPM (semiconductor driving the motor).
		(1) Check whether a valid motor number is specified in parameter No.30. (2) Check whether the standard values (see Appendix B) are specified in the current control parameters for servo control. Correct current control is possible only when the standard values are specified for the following parameters: No.70, 71, 72, 78, 79, 84, 85, 86, 87, 88, 89, 90 (3) Turn off the servo amplifier unit for about ten minutes, then release the emergency stop state. If no overcurrent alarm is issued, The IPM protection function (against overheat) may be operating. Possible causes of overheat include a high ambient temperature and a too severe motor operating condition. Check the cause. → If an overcurrent alarm is issued, go to (4). (4) Disconnect the power line from the servo amplifier unit connector. Then, release the emergency stop state. → If an overcurrent alarm is issued, the IPM protection function (against overcurrent, low control power supply voltage, overheat) may be operating or defective. Replace the IPM or servo amplifier unit. → If no overcurrent alarm is issued, go to (5). (5) Disconnect the power line from the amplifier connector. Check the insulation between the ground and each of U, V, and W. → If a short–circuit is found between the ground and U, V, or W of the motor, replace the motor. → If the insulation is satisfactory, replace the power line. (6) Connect the power line. Observe the waveform of the motor current (IR, IS) while the motor is accelerating or decelerating. See Items (8) and (9) in Appendix E, "Servo Check Board". If the motor current (IR, IS) does not exhibit a normal sine wave, replace the servo amplifier unit. (7) Check whether the waveform of the motor current (IR, IS) contains noise. → If it contains noise, take appropriate measures against noise, such as grounding the shield. → If it does not contain noise, replace the servo amplifier unit. (8) If (1) to (7) above are not the cause of the alarm, the pulse coder, command cable, or internal hardware of the CNC may be defective.	

No.	LED display	Description	Countermeasure
413	9	A DC link overvoltage alarm is issued.	This alarm is issued when the DC voltage of the main circuit power is too high.
		(1) When SVU–12 or SVU–20 is used, and a separate regenerative discharge unit is not used, check the specification to see if regenerative energy per one time does not exceed the allowable regenerative energy of the servo amplifier unit. (2) For the SVU–40 and the SVU–80, when the separate regenerative discharge resistor is not used, check whether the CX23 connector is short–circuited with a dummy connector. (3) The supply voltage for dynamic power may exceed the rated value. Check the voltage. If the voltage is too high, reduce the voltage to an appropriate level. (4) The regenerative discharge unit may not be properly connected. Check the connection. (5) The resistance of the separate regenerative discharge unit may be abnormal. Disconnect the separate regenerative discharge unit, then check the resistance. If the resistance is not within +20% of the predetermined resistance, replace the separate regenerative discharge unit. (6) If (1) to (5) are not the cause of the alarm, replace the servo amplifier unit.	
414	A	A DC link low voltage alarm is issued.	This alarm is issued when the DC voltage of the main circuit power is too low.
		(1) 190 ms or longer may pass from the time when both the *ESP of the built–in DI and the *ESP of the I/O link interface signal are canceled until the external magnetic contactor inserted into the input for motive power turns on (including the operating time of the magnetic contactor). The magnetic contactor must turn on within 100 ms. (2) The external circuit breaker may be turned off. Check the circuit breaker. (3) The supply voltage for dynamic power is lower than the rated value. Check the voltage. If the voltage is too low, increase it to an appropriate level. (4) The external magnetic contactor may not be connected properly. Check the connection. If (1) to (4) are not the cause of the alarm, replace the servo amplifier unit.	
417	8	A parameter has been specified incorrectly.	Check the following parameters: No.30: Is the specified motor type correct? No.31: Is the specified direction of rotation of the motor correct? No.106: Is the denominator of the number of pulses per single revolution of the motor 0? No.180: Is the specified reference counter capacity 0 or a negative value?
418	0	A DO alarm is issued.	Replace the servo amplifier unit.
423	E	The specified speed exceeds 32767000 detection units per second.	Re–examine the CMR and speed settings.
425	C	The cooling fan has stopped.	This alarm is issued when the fan motor built into the servo amplifier unit has failed. The fan motor is consumable. For an explanation of the replacement procedure, see Part 4, “Maintenance of Servo Amplifier Unit.”
		(1) Check that the fan is not clogged with foreign matter. (2) Check that the power connector of the fan is connected properly. (3) Replace the fan or servo amplifier unit.	
446	C	The external pulse input line is disconnected.	Connect the external pulse input signal correctly.
447	E	The velocity deviation is too high (velocity control)	Check the actual velocity. See the settings of parameter No. 136.

Overtravel alarms

No.	LED display	Description	Countermeasure
500	8	The positive stroke limit has been exceeded.	Check whether *+OT and *–OT are connected correctly. Check whether a correct move command is specified. Move the tool in the opposite direction in jog mode, then perform a reset.
501	8	The negative stroke limit has been exceeded.	

No.	LED display	Description	Countermeasure
510	8	The positive soft stroke limit has been exceeded.	Check whether appropriate values have been specified for parameters No.142 and 143. Check whether a valid move command is specified. Move the tool in the opposite direction in jog mode, then perform a reset.
511	8	The negative soft stroke limit has been exceeded.	

System alarms

No.	LED display	Description	Countermeasure
–	E	An error was detected in the RAM write/read test at power−up.	Replace the servo amplifier unit.
–	8	An error was detected in the data collation check for the battery−powered memory.	Turn the power off then back on. Then, re−enter the parameters. If this alarm recurs, replace the servo amplifier unit.
–	9	A data transfer alarm for the battery−powered memory has been issued.	Replace the servo amplifier unit.
–	8	A watchdog alarm was issued.	Turn the power off then back on. If this alarm recurs, replace the servo amplifier unit.
–	6	A checksum alarm for the control software ROM is issued.	Replace the servo amplifier unit.
–	6	A checksum alarm for the ROM that is built into the CPU is issued.	Replace the servo amplifier unit.
–	E.	An error was detected in the control circuit.	Replace the servo amplifier unit.

I/O link alarm

No.	LED display	Description	Countermeasure
–	E	A FANUC I/O Link error occurred. A servo amplifier unit connected to the line was turned off.	Turn off the power to all units connected to the line. Then, turn on the slave devices, followed by the master device.

No LED display

No.	LED display	Description	Countermeasure
–	No indicators lit	The control circuit is not operating normally.	(1) Check the 24−VDC control supply voltage. If the voltage is low, increase the voltage to an appropriate level. (2) Check whether a fuse in the servo amplifier unit has blown. If a blown fuse is found, replace it, following the procedure described in Part 4 “Maintenance of Servo Amplifier Unit.” If (1) and (2) are not the cause, replace the servo amplifier.